

Regression Report

My dataset is Advertising dataset. It has 200 observations with 4 features.
I did linear regression on it using Closed Form Solution then performed Polynomial Regression with different degrees

Model Degree	Validation R^2	Test R^2	Observation
1 (Linear)	0.837	0.907	Simple linear model
2	0.88	0.95	Best generalization; captures some non-linear patterns
3	0.81	0.90	Slight overfitting; R^2 drops on validation/test
4	0.84	0.70	Overfitting stronger; validation/test R^2 lower
5	0.74	-1.73	Overfitting very badly! R^2 is negative

Observations:

- Linear regression is performing good, But polynomial regression with degree 2 is generalizing the model the best.
- Higher Degrees like 3,4,5 are tending to overfit

Lasso and Ridge Regularization:

I applied Lasso and Ridge Regularization on the linear model with different alpha values

Regularization	Alpha	Validation R^2	Test R^2	Observation
Lasso (L1)	0.01	0.837	0.907	Small effect; coefficients slightly shrunk
Lasso (L1)	0.1	0.844	0.908	Slight more effect
Lasso (L1)	1	0.827	0.839	Many coefficients shrink; validation R^2 decreased
Lasso (L1)	10	-0.051	-0.05	Over-regularization; model predicts near-zero $\rightarrow R^2$ negative
Ridge (L2)	0.01	0.837	0.907	Small shrinkage; stable

Ridge (L2)	0.1	0.837	0.907	Slightly reduced coefficients; stable
Ridge (L2)	1	0.839	0.907	Stronger shrinkage; R^2 still positive
Ridge (L2)	10	0.850	0.903	Heavy shrinkage; R^2 slightly lower but still reasonable

Observations:

- Lasso is sensitive to large alpha, it is underfitting severely (negative R^2)
- Ridge is more stable, it does smooth shrinkage

Gradient Descent Convergence:

I compared three variants

- Full-batch Gradient Descent (all samples each update)
- Mini-batch Gradient Descent (batch size 16)
- Stochastic Gradient Descent (SGD) (batch size 1)

Observations:

- Loss curves for all three methods were almost identical.
- Because my dataset is small (few features, few samples) and linear regression loss function is convex and simple.
- Differences would appear for large/high-dimensional datasets or non-linear models.

Loss Curve Summary:

- Full-batch has smooth convergence
- Mini-batch is very slightly noisy
- SGD has slight fluctuation per epoch
- All converge to the same global minimum.

Overall Conclusion:

1. **Polynomial regression** can improve R^2 over linear regression, but higher degrees (3,4,5) may overfit small datasets.
2. **Regularization** helps prevent overfitting:
 - a. L1 (Lasso) → sparse coefficients, sensitive to large alpha
 - b. L2 (Ridge) → smooth shrinkage, more stable
3. **Gradient descent variants** perform similarly on small datasets: differences become relevant only on large datasets or high-dimensional models.